CS-104 Retrieval Quiz (ANSWER KEY)

Fall 2026 · Week 7, Monday

Name: _____

Closed book, no devices. For questions 1–3, circle the one best answer. For question 4, write your code in the space provided (no algorithm, documentation, or test cases needed).

1.

What do the two print statements output?

```
import numpy as np
values = [4, 5]
arr = np.array(values)
print(values * 3)
print(arr * 3)
```

a) `[12, 15]`, then `[12 15]`

b) `[4, 5, 4, 5, 4, 5]`**, then** `[12 15]` **[correct]**

c) `[4, 5, 4, 5, 4, 5]`, then `[4 5 4 5 4 5]`

d) A `TypeError` for both

2.

Which call produces the array `[0. 0.25 0.5 0.75 1.]`?

a) `np.arange(0, 1, 0.25)`

b) `np.linspace(0, 1, 4)`

c) `np.arange(0, 1, 5)`

d) `np.linspace(0, 1, 5)` **[correct]**

3.

What does this code print?

```
import numpy as np
x = np.array([0, 1, 2, 3])
y = x ** 2
print(y)
```

a) `[0 1 4 9]` **[correct]**

b) `[0 2 4 6]`

c) `[0, 1, 2, 3, 0, 1, 2, 3]`

d) It raises a `TypeError`

4.

A radioactive sample decays over time. The amount left after `t` years is

N = 100 × e^(-0.5 t)

Write a code segment that uses NumPy to create 50 evenly spaced times from 0 to 8 years, computes the amount left at each time (use `np.exp`), and uses Matplotlib to plot amount versus time with the axis labels `Time (years)` and `Amount (g)` and a title.

Sample solution:

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0, 8, 50)
n = 100 * np.exp(-0.5 * t)
plt.plot(t, n)
plt.xlabel("Time (years)")
plt.ylabel("Amount (g)")
plt.title("Radioactive Decay")
plt.show()
```